

# STM32 CubeMX

## 1. Description

### 1.1. Project

|                 |                    |
|-----------------|--------------------|
| Project Name    | tractive-battery   |
| Board Name      | custom             |
| Generated with: | STM32CubeMX 6.15.0 |
| Date            | 11/15/2025         |

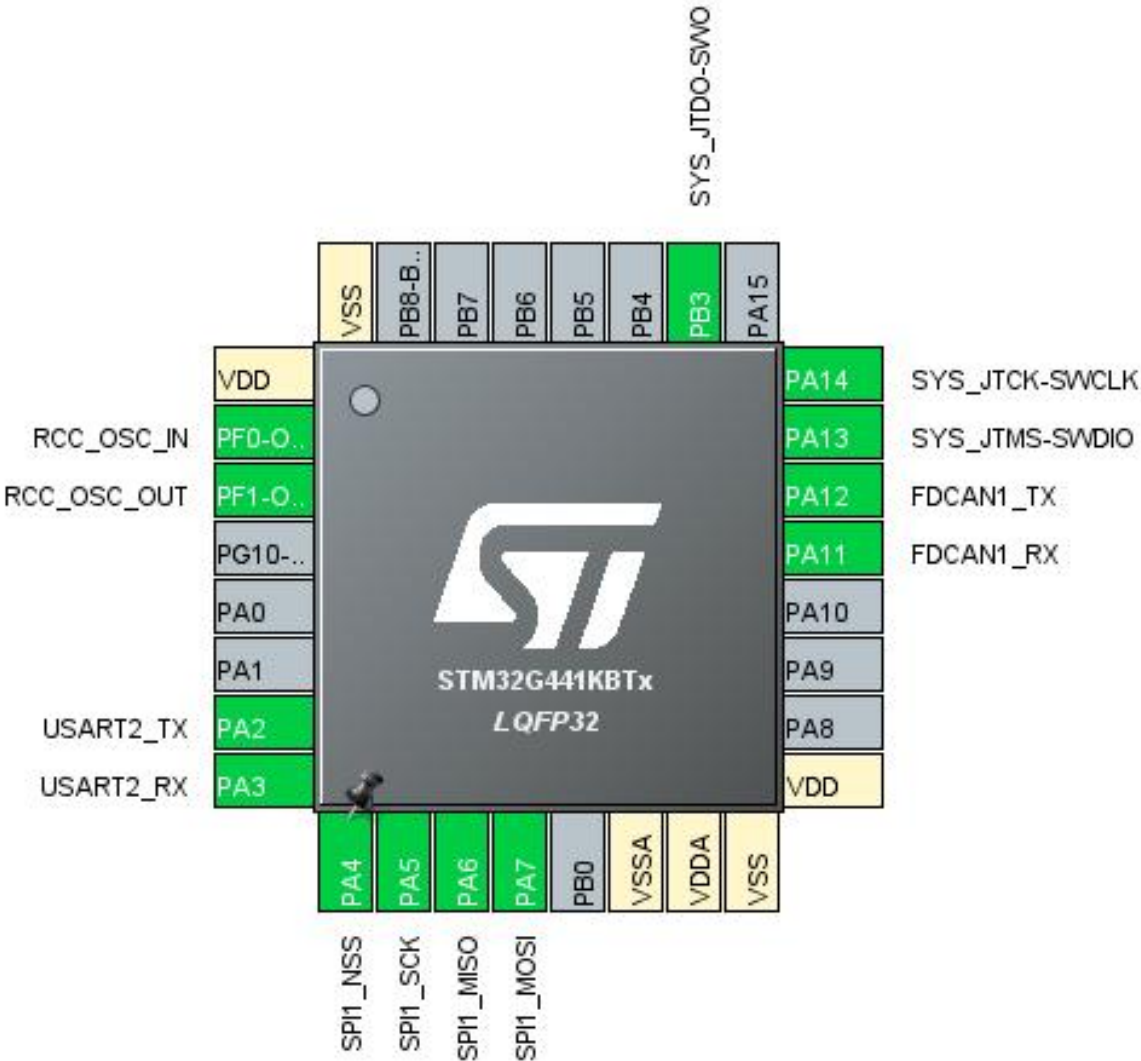
### 1.2. MCU

|                |               |
|----------------|---------------|
| MCU Series     | STM32G4       |
| MCU Line       | STM32G4x1     |
| MCU name       | STM32G441KBTx |
| MCU Package    | LQFP32        |
| MCU Pin number | 32            |

### 1.3. Core(s) information

|         |               |
|---------|---------------|
| Core(s) | ARM Cortex-M4 |
|---------|---------------|

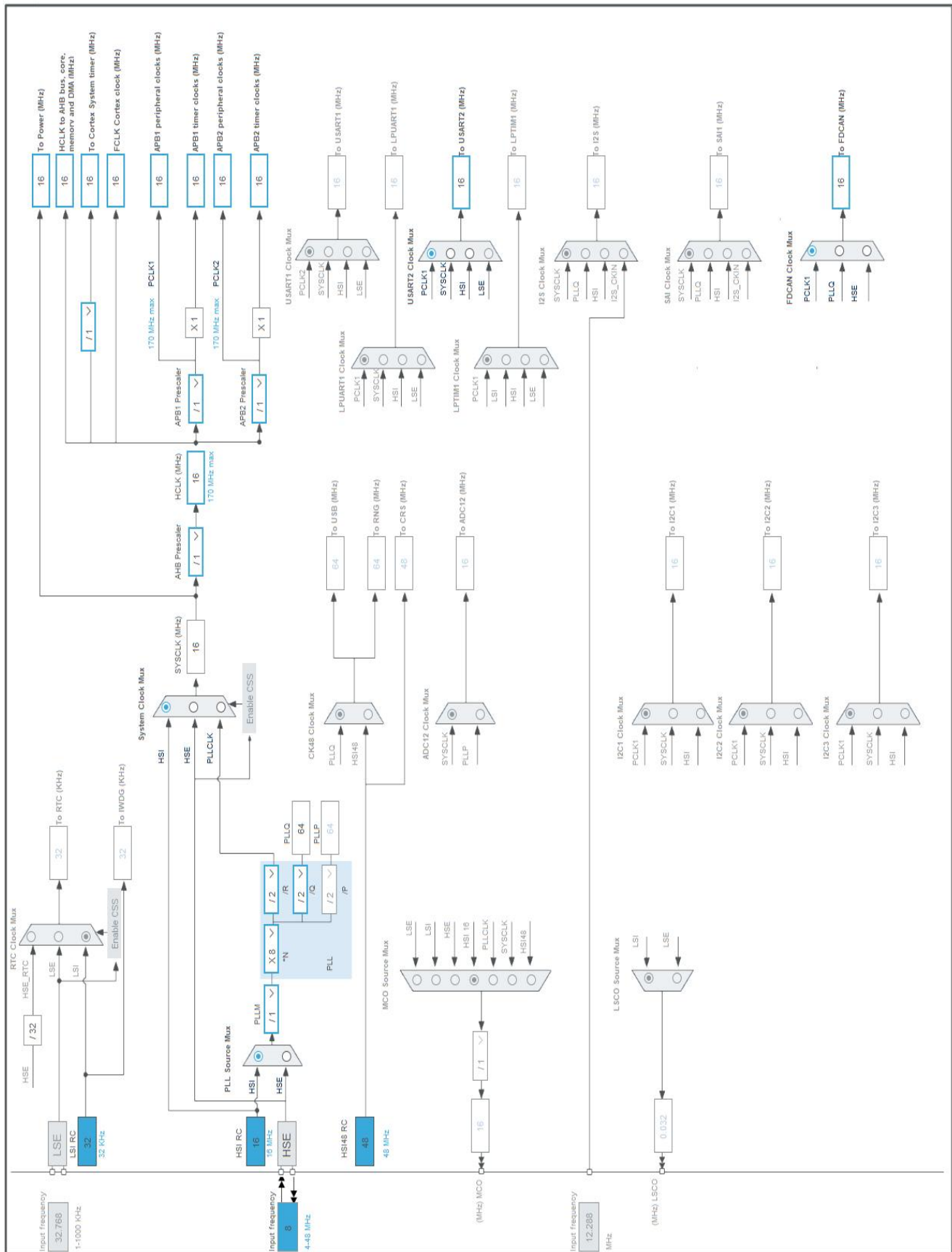
## 2. Pinout Configuration



### 3. Pins Configuration

| Pin Number<br>LQFP32 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label |
|----------------------|---------------------------------------|----------|--------------------------|-------|
| 1                    | VDD                                   | Power    |                          |       |
| 2                    | PF0-OSC_IN                            | I/O      | RCC_OSC_IN               |       |
| 3                    | PF1-OSC_OUT                           | I/O      | RCC_OSC_OUT              |       |
| 7                    | PA2                                   | I/O      | USART2_TX                |       |
| 8                    | PA3                                   | I/O      | USART2_RX                |       |
| 9                    | PA4                                   | I/O      | SPI1_NSS                 |       |
| 10                   | PA5                                   | I/O      | SPI1_SCK                 |       |
| 11                   | PA6                                   | I/O      | SPI1_MISO                |       |
| 12                   | PA7                                   | I/O      | SPI1_MOSI                |       |
| 14                   | VSSA                                  | Power    |                          |       |
| 15                   | VDDA                                  | Power    |                          |       |
| 16                   | VSS                                   | Power    |                          |       |
| 17                   | VDD                                   | Power    |                          |       |
| 21                   | PA11                                  | I/O      | FDCAN1_RX                |       |
| 22                   | PA12                                  | I/O      | FDCAN1_TX                |       |
| 23                   | PA13                                  | I/O      | SYS_JTMS-SWDIO           |       |
| 24                   | PA14                                  | I/O      | SYS_JTCK-SWCLK           |       |
| 26                   | PB3                                   | I/O      | SYS_JTDO-SWO             |       |
| 32                   | VSS                                   | Power    |                          |       |

## 4. Clock Tree Configuration



## 1. Power Consumption Calculator report

### 1.1. Microcontroller Selection

|           |               |
|-----------|---------------|
| Series    | STM32G4       |
| Line      | STM32G4x1     |
| MCU       | STM32G441KBTx |
| Datasheet | DS12589 Rev0  |

### 1.2. Parameter Selection

|             |     |
|-------------|-----|
| Temperature | 25  |
| Vdd         | 3.0 |

### 1.3. Battery Selection

|                   |                 |
|-------------------|-----------------|
| Battery           | Li-SOCL2(A3400) |
| Capacity          | 3400.0 mAh      |
| Self Discharge    | 0.08 %/month    |
| Nominal Voltage   | 3.6 V           |
| Max Cont Current  | 100.0 mA        |
| Max Pulse Current | 200.0 mA        |
| Cells in series   | 1               |
| Cells in parallel | 1               |

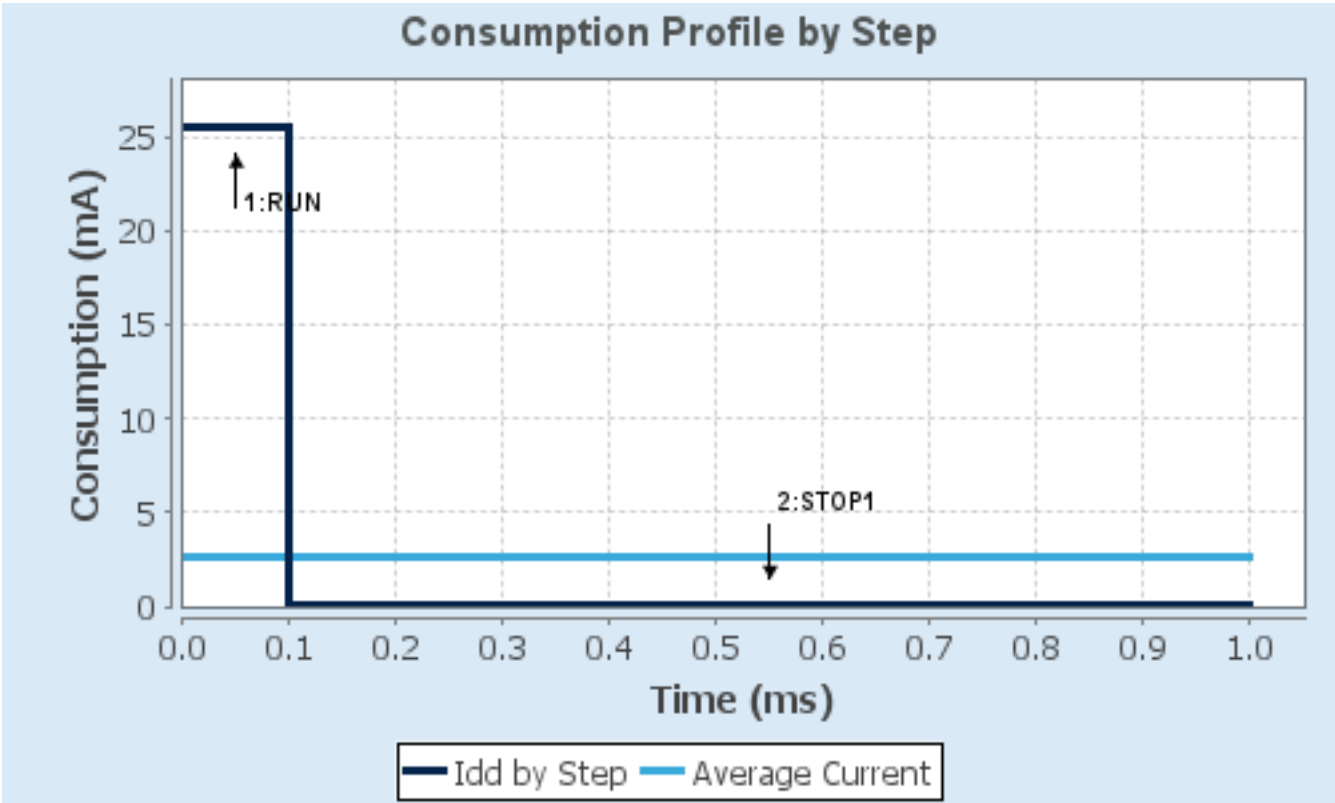
## 1.4. Sequence

|                               |              |                |
|-------------------------------|--------------|----------------|
| <b>Step</b>                   | Step1        | Step2          |
| <b>Mode</b>                   | RUN          | STOP1          |
| <b>Vdd</b>                    | 3.0          | 3.0            |
| <b>Voltage Source</b>         | Battery      | Battery        |
| <b>Range</b>                  | Range1-Boost | NoRange        |
| <b>Fetch Type</b>             | FLASH/ART    | NA             |
| <b>CPU Frequency</b>          | 170 MHz      | 0 Hz           |
| <b>Clock Configuration</b>    | HSE BYP PLL  | ALL CLOCKS OFF |
| <b>Clock Source Frequency</b> | 4 MHz        | 0 Hz           |
| <b>Peripherals</b>            |              |                |
| <b>Additional Cons.</b>       | 0 mA         | 0 mA           |
| <b>Average Current</b>        | 25.5 mA      | 59 $\mu$ A     |
| <b>Duration</b>               | 0.1 ms       | 0.9 ms         |
| <b>DMIPS</b>                  | 213.0        | 0.0            |
| <b>Ta Max</b>                 | 124.19       | 129.99         |
| <b>Category</b>               | In DS Table  | In DS Table    |

## 1.5. Results

|               |                               |                 |             |
|---------------|-------------------------------|-----------------|-------------|
| Sequence Time | 1 ms                          | Average Current | 2.6 mA      |
| Battery Life  | 1 month, 23 days,<br>22 hours | Average DMIPS   | 212.5 DMIPS |

## 1.6. Chart





## 2. Software Project

### 2.1. Project Settings

| Name                              | Value                                         |
|-----------------------------------|-----------------------------------------------|
| Project Name                      | tractive-battery                              |
| Project Folder                    | C:\Users\Daniel Du\Desktop\Formula Slug\fs-4- |
| Toolchain / IDE                   | EWARM V8.50                                   |
| Firmware Package Name and Version | STM32Cube FW_G4 V1.6.1                        |
| Application Structure             | Advanced                                      |
| Generate Under Root               | No                                            |
| Do not generate the main()        | No                                            |
| Minimum Heap Size                 | 0x200                                         |
| Minimum Stack Size                | 0x400                                         |

### 2.2. Code Generation Settings

| Name                                                            | Value                                           |
|-----------------------------------------------------------------|-------------------------------------------------|
| STM32Cube MCU packages and embedded software                    | Copy all used libraries into the project folder |
| Generate peripheral initialization as a pair of '.c/.h' files   | No                                              |
| Backup previously generated files when re-generating            | No                                              |
| Keep User Code when re-generating                               | Yes                                             |
| Delete previously generated files when not re-generated         | Yes                                             |
| Set all free pins as analog (to optimize the power consumption) | No                                              |
| Enable Full Assert                                              | No                                              |

### 2.3. Advanced Settings - Generated Function Calls

| Rank | Function Name       | Peripheral Instance Name |
|------|---------------------|--------------------------|
| 1    | SystemClock_Config  | RCC                      |
| 2    | MX_GPIO_Init        | GPIO                     |
| 3    | MX_FDCAN1_Init      | FDCAN1                   |
| 4    | MX_SPI1_Init        | SPI1                     |
| 5    | MX_USART2_UART_Init | USART2                   |

## 3. Peripherals and Middlewares Configuration

### 3.1. FDCAN1

**mode: Activated**

#### 3.1.1. Parameter Settings:

##### Basic Parameters:

|                         |                          |
|-------------------------|--------------------------|
| Clock Divider           | Divide kernel clock by 1 |
| Frame Format            | Classic mode             |
| Mode                    | Normal mode              |
| Auto Retransmission     | Disable                  |
| Transmit Pause          | Disable                  |
| Protocol Exception      | Disable                  |
| Nominal Sync Jump Width | 1                        |
| Data Prescaler          | 1                        |
| Data Sync Jump Width    | 1                        |
| Data Time Seg1          | 1                        |
| Data Time Seg2          | 1                        |
| Std Filters Nbr         | 0                        |
| Ext Filters Nbr         | 0                        |
| Tx Fifo Queue Mode      | FIFO mode                |

##### Bit Timings Parameters:

|                          |                 |
|--------------------------|-----------------|
| Nominal Prescaler        | 16              |
| Nominal Time Quantum     | <b>1000.0 *</b> |
| Nominal Time Seg1        | 1               |
| Nominal Time Seg2        | 1               |
| Nominal Time for one Bit | <b>3000 *</b>   |
| Nominal Baud Rate        | <b>333333 *</b> |

### 3.2. RCC

**High Speed Clock (HSE): Crystal/Ceramic Resonator**

#### 3.2.1. Parameter Settings:

##### System Parameters:

|                   |                    |
|-------------------|--------------------|
| VDD voltage (V)   | 3.3                |
| Instruction Cache | Enabled            |
| Prefetch Buffer   | Disabled           |
| Data Cache        | Enabled            |
| Flash Latency(WS) | 0 WS (1 CPU cycle) |

**RCC Parameters:**

|                                |      |
|--------------------------------|------|
| HSI Calibration Value          | 64   |
| HSE Startup Timeout Value (ms) | 100  |
| LSE Startup Timeout Value (ms) | 5000 |

**Power Parameters:**

|                               |                                 |
|-------------------------------|---------------------------------|
| Power Regulator Voltage Scale | Power Regulator Voltage Scale 1 |
|-------------------------------|---------------------------------|

**Peripherals Clock Configuration:**

|                                              |      |
|----------------------------------------------|------|
| Generate the peripherals clock configuration | TRUE |
|----------------------------------------------|------|

### 3.3. SPI1

**Mode: Full-Duplex Master**

**Hardware NSS Signal: Hardware NSS Output Signal**

#### 3.3.1. Parameter Settings:

**Basic Parameters:**

|              |           |
|--------------|-----------|
| Frame Format | Motorola  |
| Data Size    | 4 Bits    |
| First Bit    | MSB First |

**Clock Parameters:**

|                           |                      |
|---------------------------|----------------------|
| Prescaler (for Baud Rate) | 2                    |
| Baud Rate                 | <b>8.0 MBits/s *</b> |
| Clock Polarity (CPOL)     | Low                  |
| Clock Phase (CPHA)        | 1 Edge               |

**Advanced Parameters:**

|                 |                 |
|-----------------|-----------------|
| CRC Calculation | Disabled        |
| NSSP Mode       | Enabled         |
| NSS Signal Type | Output Hardware |

### 3.4. SYS

**Debug: Trace Asynchronous Sw**

**Timebase Source: SysTick**

**mode: save power of non-active UCPD - deactive Dead Battery pull-up**

### 3.5. USART2

**Mode: Asynchronous**

### 3.5.1. Parameter Settings:

#### **Basic Parameters:**

|             |                           |
|-------------|---------------------------|
| Baud Rate   | 115200                    |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

#### **Advanced Parameters:**

|                  |                             |
|------------------|-----------------------------|
| Data Direction   | Receive and Transmit        |
| Over Sampling    | 16 Samples                  |
| Single Sample    | Disable                     |
| ClockPrescaler   | 1                           |
| Fifo Mode        | Disable                     |
| Txfifo Threshold | 1 eighth full configuration |
| Rxfifo Threshold | 1 eighth full configuration |

#### **Advanced Features:**

|                               |         |
|-------------------------------|---------|
| Auto Baudrate                 | Disable |
| TX Pin Active Level Inversion | Disable |
| RX Pin Active Level Inversion | Disable |
| Data Inversion                | Disable |
| TX and RX Pins Swapping       | Disable |
| Overrun                       | Enable  |
| DMA on RX Error               | Enable  |
| MSB First                     | Disable |

**\* User modified value**

## 4. System Configuration

### 4.1. GPIO configuration

| IP     | Pin         | Signal         | GPIO mode                    | GPIO pull/up pull down      | Max Speed | User Label |
|--------|-------------|----------------|------------------------------|-----------------------------|-----------|------------|
| FDCAN1 | PA11        | FDCAN1_RX      | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|        | PA12        | FDCAN1_TX      | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
| RCC    | PF0-OSC_IN  | RCC_OSC_IN     | n/a                          | n/a                         | n/a       |            |
|        | PF1-OSC_OUT | RCC_OSC_OUT    | n/a                          | n/a                         | n/a       |            |
| SPI1   | PA4         | SPI1_NSS       | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|        | PA5         | SPI1_SCK       | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|        | PA6         | SPI1_MISO      | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|        | PA7         | SPI1_MOSI      | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
| SYS    | PA13        | SYS_JTMS-SWDIO | n/a                          | n/a                         | n/a       |            |
|        | PA14        | SYS_JTCK-SWCLK | n/a                          | n/a                         | n/a       |            |
|        | PB3         | SYS_JTDO-SWO   | n/a                          | n/a                         | n/a       |            |
| USART2 | PA2         | USART2_TX      | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|        | PA3         | USART2_RX      | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |

### 4.2. DMA configuration

nothing configured in DMA service

### 4.3. NVIC configuration

#### 4.3.1. NVIC

| Interrupt Table                                                         | Enable | Preenmption Priority | SubPriority |
|-------------------------------------------------------------------------|--------|----------------------|-------------|
| Non maskable interrupt                                                  | true   | 0                    | 0           |
| Hard fault interrupt                                                    | true   | 0                    | 0           |
| Memory management fault                                                 | true   | 0                    | 0           |
| Prefetch fault, memory access fault                                     | true   | 0                    | 0           |
| Undefined instruction or illegal state                                  | true   | 0                    | 0           |
| System service call via SWI instruction                                 | true   | 0                    | 0           |
| Debug monitor                                                           | true   | 0                    | 0           |
| Pendable request for system service                                     | true   | 0                    | 0           |
| System tick timer                                                       | true   | 15                   | 0           |
| PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/38/39/40/41    | unused |                      |             |
| Flash global interrupt                                                  | unused |                      |             |
| RCC global interrupt                                                    | unused |                      |             |
| FDCAN1 interrupt 0                                                      | unused |                      |             |
| FDCAN1 interrupt 1                                                      | unused |                      |             |
| SPI1 global interrupt                                                   | unused |                      |             |
| USART2 global interrupt / USART2 wake-up interrupt through EXTI line 26 | unused |                      |             |
| FPU global interrupt                                                    | unused |                      |             |

#### 4.3.2. NVIC Code generation

| Enabled interrupt Table                 | Select for init sequence ordering | Generate IRQ handler | Call HAL handler |
|-----------------------------------------|-----------------------------------|----------------------|------------------|
| Non maskable interrupt                  | false                             | true                 | false            |
| Hard fault interrupt                    | false                             | true                 | false            |
| Memory management fault                 | false                             | true                 | false            |
| Prefetch fault, memory access fault     | false                             | true                 | false            |
| Undefined instruction or illegal state  | false                             | true                 | false            |
| System service call via SWI instruction | false                             | true                 | false            |
| Debug monitor                           | false                             | true                 | false            |
| Pendable request for system service     | false                             | true                 | false            |
| System tick timer                       | false                             | true                 | true             |

\* User modified value

## 5. System Views

### 5.1. Category view

#### 5.1.1. Current

| Middleware  |        |        |              |            |          |           |           |
|-------------|--------|--------|--------------|------------|----------|-----------|-----------|
| System Core | Analog | Timers | Connectivity | Multimedia | Security | Computing | Utilities |
| DMA         |        |        | FDCAH1 ✓     |            |          |           |           |
| GPIO ✓      |        |        | SPI1 ✓       |            |          |           |           |
| IIVIC ✓     |        |        | USART2 ✓     |            |          |           |           |
| RCC ✓       |        |        |              |            |          |           |           |
| SYS ✓       |        |        |              |            |          |           |           |

## 6. Docs & Resources

| Type                    | Link                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BSDL files              | <a href="https://www.st.com/resource/en/bsdl_model/stm32g4_bsd1.zip">https://www.st.com/resource/en/bsdl_model/stm32g4_bsd1.zip</a>                                                                                                                                                                                                     |
| IBIS models             | <a href="https://www.st.com/resource/en/ibis_model/stm32g4_ibis.zip">https://www.st.com/resource/en/ibis_model/stm32g4_ibis.zip</a>                                                                                                                                                                                                     |
| System View Description | <a href="https://www.st.com/resource/en/svd/stm32g4_svd.zip">https://www.st.com/resource/en/svd/stm32g4_svd.zip</a>                                                                                                                                                                                                                     |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf</a>                                                                                                                             |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf">https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf</a>                                                                                                                                                     |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf">https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf</a>                                                                                                                               |
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| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf</a>                                                                                                                               |
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| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32g4-series-product-overview.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32g4-series-product-overview.pdf</a>                                                                                                         |
| Brochures               | <a href="https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf">https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf</a>                                                                                                                                     |
| Brochures               | <a href="https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf">https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf</a>                                                                                                                                   |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32g4.pdf">https://www.st.com/resource/en/flyer/flstm32g4.pdf</a>                                                                                                                                                                                                                     |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32nucleo.pdf">https://www.st.com/resource/en/flyer/flstm32nucleo.pdf</a>                                                                                                                                                                                                             |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32trust.pdf">https://www.st.com/resource/en/flyer/flstm32trust.pdf</a>                                                                                                                                                                                                               |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstpfc11120.pdf">https://www.st.com/resource/en/flyer/flstpfc11120.pdf</a>                                                                                                                                                                                                               |
| Security Bulletin       | <a href="https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf</a> |



Application Notes [https://www.st.com/resource/en/application\\_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf)

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Application Notes [https://www.st.com/resource/en/application\\_note/an3155-uart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3155-uart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf)

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Application Notes [https://www.st.com/resource/en/application\\_note/an4232-getting-started-with-analog-comparators-for-stm32f3-series-and-stm32g4-series-devices-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4232-getting-started-with-analog-comparators-for-stm32f3-series-and-stm32g4-series-devices-stmicroelectronics.pdf)

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Application Notes [https://www.st.com/resource/en/application\\_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf)

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- Application Notes [https://www.st.com/resource/en/application\\_note/an5093-getting-started-with-stm32g4-series--hardware-development-boards-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5093-getting-started-with-stm32g4-series--hardware-development-boards-stmicroelectronics.pdf)
- Application Notes [https://www.st.com/resource/en/application\\_note/an5306-operational-amplifier-opamp-usage-in-stm32g4-series-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5306-operational-amplifier-opamp-usage-in-stm32g4-series-stmicroelectronics.pdf)
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| for related Tools<br>& Software                      | <a href="#">implementation-with-the-fmac-using-stm32cubeg4-mcu-package-stmicroelectronics.pdf</a>                                                                                                                                                                                                                                                                                         |
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| Datasheet                     | <a href="https://www.st.com/resource/en/datasheet/dm00607954.pdf">https://www.st.com/resource/en/datasheet/dm00607954.pdf</a>                                                                                                                                                                                                                                 |
| Programming<br>Manuals        | <a href="https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf">https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf</a>                                                                                 |
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| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf</a>                       |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf</a>                         |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf</a> |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf</a>                   |
| User Manuals                  | <a href="https://www.st.com/resource/en/user_manual/um3167-stm32g4-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf">https://www.st.com/resource/en/user_manual/um3167-stm32g4-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf</a>                                                             |